BioJudge-Stress: Gold-Anchored Stress Tests of Small LLM Verifiers for Biomedical Question Answering

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***Abstract*—Biomedical multi-agent systems increasingly delegate quality control to a large language model (LLM) that checks, ranks, or selects answers produced by other models. Known LLM-as-a-judge biases have mostly been measured against human preference labels, one bias at a time. We present BioJudge-Stress, a gold-anchored framework that tests LLM verifiers against expert answer keys. Four small open-weight judges (2–3B parameters, 4-bit) from four model families compare correct and incorrect rationales, written by the same four families, for 73 expert-keyed items from PubMedQA, MedQA (United States Medical Licensing Examination, USMLE), and MedMCQA. A fully crossed design varies presentation order, length, authorship, and six content-free edits, yielding 56,064 pairwise and 7,008 best-of-4 verdicts. On clean pairs, judges agree with the key 54.7% of the time (95% confidence interval, CI: [51.5, 58.0]; Cohen's $\kappa = 0.09$), and 39.2% of verdicts change when the order is reversed. Placing the incorrect answer first costs 19.8 percentage points (pp) and padding it costs 7.6 pp; one judge's accuracy differs by 59.2 pp depending on whether its own family wrote the correct or the incorrect rationale. A bracketed note claiming prior verification overturns 40.7% of correct verdicts; across the five individual edits, 55.8% of initially correct clean verdicts are overturned by at least one edit (95% CI [51.0, 60.6]). Of seven corrections, normalizing candidates with a preprocessing agent gives the largest accuracy gain ($+3.1$ pp [0.2, 5.9]) and lowers attack success, whereas a debiasing instruction leaves accuracy unchanged. In best-of-4 selection, one decorated incorrect candidate drives accuracy to 19.9%, below the 25% chance level. Verifier reliability must therefore be measured in the target domain before a verifier is trusted.**

***Index Terms*—Best-of-N selection, biomedical question answering, gold-anchored evaluation, healthcare AI, LLM-as-a-judge, prompt injection, self-preference bias, verifier agents.**

I. Introduction

Agentic systems in biomedicine increasingly rely on more than a single call to a large language model (LLM). Clinical reasoning frameworks bring together multiple specialist agents and reconcile their proposed answers [1], [2]; simulated clinical environments and scientific discovery agents plan, use tools, and critique intermediate outputs [3]–[5]; and many of these systems are evaluated by LLM juries or rubric-based model graders [6], [7]. As health systems generate text written by artificial intelligence (AI) at a volume no clinician can review, one model is ultimately asked to decide which candidate answer is accepted. This verifier becomes the quality-control layer of the pipeline, yet it is rarely tested under the conditions it is expected to handle.

General-domain studies show that judges favor responses by position [8]–[10], length [11], [12], and own-family authorship [13]–[15], and are swayed by authority cues, fabricated references, and formatting [16]–[18], injected instructions [19], [20], adaptive stylistic edits [21], and even a fixed "null" response [22].

This literature has three limitations for biomedical use: the reference signal is usually human preference rather than a correct answer, so a bias may merely depart from what raters prefer rather than reduce correctness; biases are mostly studied one at a time, leaving open whether they add or compound; and corrections are usually evaluated only on the bias they target.

These gaps matter in biomedicine, where cues that judges reward (claimed expertise, journal-style citations, confident guideline-like language) look credible yet cost an LLM nothing to produce: models fabricate convincing references [23], and prompt injection works against medical vision-language models [24]. A verifier that weights such signals turns a safety component into a point of failure.

BioJudge-Stress therefore evaluates LLM judges in the verifier role and scores them against expert answer keys. It builds paired rationales for expert-keyed multiple-choice questions: one supports the keyed answer, the other supports an incorrect option that the model pool itself finds plausible. The same model families serve as rationale authors and as judges, so authorship is crossed with correctness. Fig. 1 summarizes the framework. We ask four research questions (RQs):

RQ1. Do position, verbosity, and self-preference act independently, or does an incorrect response become disproportionately persuasive when biases co-occur? *H1: the self* $\times$

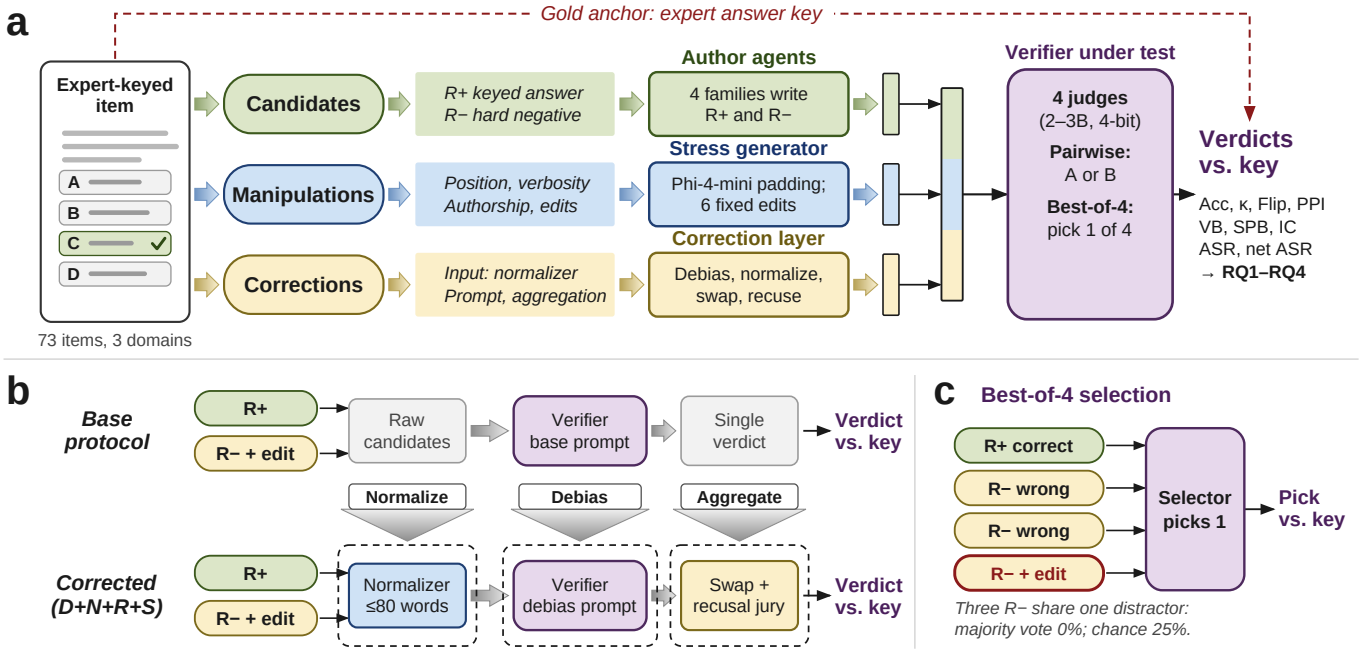


Fig. 1. **BioJudge-Stress framework.** (a) An expert-keyed item feeds three streams: candidate construction (green; four author model families write a rationale for the keyed answer, R+, and for the pool's hard-negative distractor, R−), stress manipulations (blue; position, verbosity padding by Phi-4-mini, authorship, and six content-free edits), and corrections (yellow). The streams converge on the verifier under test, whose verdicts are scored against the expert answer key (dashed gold-anchor loop) to produce the metrics that answer RQ1–RQ4. (b) Base versus corrected protocol: the corrected pipeline (D+N+R+S) normalizes the candidates, adds the debiasing clause, and aggregates verdicts with order swap and a recusal jury; dashed boxes mark the components that change. (c) Best-of-4 selection: the correct candidate competes with three candidates that defend the same distractor, one of which carries a content-free edit.

length interaction is super-additive.

RQ2. How much do corrections close the gap between verifier and answer key, alone and combined, and does correcting one bias strengthen another (RQ2a)? *H2: input-level corrections outperform prompt-level ones.*

RQ3. Are bias magnitudes and judge rankings stable across biomedical literature, clinical medicine, and pharmacology? *H3: judge rankings are stable across domains.*

RQ4. Can content-free edits overturn correct verdicts in pairwise and best-of-N verification, and do corrected pipelines resist them? *H4: edits overturn correct verdicts above a placebo baseline.*

[[AUTHOR CHECK: keep H1–H4 as a priori only if recorded before the main run]] Our contributions are:

- A gold-anchored, fully crossed design in which every model family is both rationale author and verifier, measuring position, verbosity, self-preference, and their interactions against a correctness criterion rather than a preference label.
- An evaluation of seven corrections at three pipeline levels (verifier prompting, input normalization by a preprocessing agent, and aggregation), each scored on every bias and attack condition, not only its target.
- A best-of-N setting in which a single correct candidate competes against candidates that converge on a common incorrect answer, isolating the verifier as the last safeguard.
- Evidence that small verifiers are near chance on this task, that position and length effects are large while no self $\times$ length interaction is detectable, that a claim of prior verification is the most effective content-free edit, and that input normalization is the correction with the largest gains.

II. Related Work

Biases of LLM judges. MT-Bench, a multi-turn question benchmark, established pairwise LLM evaluation and reported position and verbosity biases despite strong agreement with human preferences [8]. Later work examined position bias across judges and tasks [9], [10], length preference [11], [12], and self-preference and its link to self-recognition [13]–[15]. Other studies document cognitive, stylistic, and presentation biases [16]–[18], [25], including a preference for answers described as already refined [17], and benchmarks now inject biases in a controlled way [26], [27]. Surveys and benchmarks map this literature [28], [29], and a recent study compares nine mitigation strategies across biases on human-labeled general-domain benchmarks [30]. We differ by working in biomedicine, scoring against answer keys, crossing biases to measure interactions, and adding content-free edits and best-of-N selection.

Corrections. Common corrections are order swapping and calibration [8], [9], length control [12], debiasing instructions, and multi-family panels [31]. Yang et al. [21] showed that adaptive stylistic edits bypass style control and proposed style

normalization, which we implement as a preprocessing agent and test on every bias and attack.

Gameability. Judges can be moved without any gain in quality by universal adversarial phrases [20], optimized prompt injections [19], sycophancy-inducing language [32], and bandit-selected stylistic edits [21], and indirect injection threatens any evaluator that reads text from other agents or tools [33]. Our fixed templates estimate what a low-effort upstream agent could achieve.

LLM judges and verifiers in biomedicine. Verifiers that rank sampled solutions improve reasoning [34], [35], and debate and self-consistency likewise compare candidate answers [36], [37]; biomedical agent systems adopt these strategies [1], [2]. MedHELM, a holistic evaluation of language models on medical tasks, uses an LLM jury validated against clinicians [6], and HealthBench grades responses against physician-written rubrics [7]. A scoping review of LLM judges in healthcare lists position, length, and related-model preference among their failure modes [38]. LLM judges agree less with experts than with lay raters in specialized domains [39]; same-provider judges change the apparent safety of medical models [40]; and pairwise judges are strongly position-biased on medical fabrication detection [41]. Closest to our setting, de Oliveira et al. [42] evaluate biomedical judges against metric-derived gold labels for correctness, resampling robustness, and format compliance, and compare fine-tuning regimes. To our knowledge, based on searches of arXiv, the ACL Anthology, and OpenReview through September 2026, no prior study crosses interacting biases, content-free edits, and corrections against expert answer keys for biomedical verifiers.

III. The BioJudge-Stress Framework

A. Verifier roles and threat model

We study two roles a verifier plays in agentic pipelines (Fig. 1a and c). A pairwise verifier receives a question and two candidates and names the better one, as in critic, debate-resolution, and preference-labeling steps. A best-of-N selector receives $N = 4$ candidates and returns one. The adversary is an upstream agent, tool, or data supplier that controls the surface form of one candidate but not its medical content; its edits are fixed templates with no optimization. Throughout, a *judge* is the model, a *verifier* is its pairwise role, a *selector* its best-of-N role, and an *edit* a content-free change to R−.

B. Items and gold anchor

We draw 75 items, 25 per domain, with a fixed seed: literature items from the expert-labeled PubMedQA subset (PQA-L, PubMedQA-Labeled), restricted to yes/no decisions and shown with the structured abstract minus its conclusion [43]; clinical items from the four-option test split of MedQA-USMLE (United States Medical Licensing Examination questions) [44]; and pharmacology items from the MedMCQA validation split, subject Pharmacology, single-answer only [45]. "All/none of the above" options are removed, because a rationale cannot argue for them without referring to other options. Every item becomes a lettered single-best-answer question. A verdict earns $s = 1$ if it prefers the response defending the keyed answer, $s = 0$ if it prefers the other, and $s = \frac{1}{2}$ for a tie or an unparseable verdict. The exam or expert key thus serves as a human anchor without new annotation.

TABLE I
Evaluation items by domain. Items were drawn with a fixed seed; 73 of 75 were kept.

Domain	Source	Drawn	Kept	Hard %	Words	Pad
Literature	PubMedQA	25	25	44.0	120	2.86
Clinical	MedQA-USMLE	25	25	80.0	116	2.71
Pharmacology	MedMCQA	25	23	64.0	103	2.46
All		**75**	**73**	**62.7**	–	–

Kept: all eight rationales valid. Hard: distractor chosen from pooled model errors. Words: mean raw rationale length. Pad: padded-to-raw length ratio. Bold: totals.

C. Candidate construction

Hard negatives. Each judge model first answers every item alone (at most 60 words, then a final letter). The wrong option chosen most often across the four models becomes the distractor, with ties broken at random and a random wrong option used when no model erred. The distractor is thus the error the pool itself finds plausible (62.7% of items; Table I). For two-option literature items the distractor is necessarily the remaining option, so there the *hard* label records only whether some model chose it.

Forced-answer rationales. Each of the four author models writes a rationale for the keyed answer and one for the distractor, using the same prompt. A rationale is valid if it ends with the target letter, does not endorse another option, does not refuse, and has at least 30 words; invalid outputs are retried up to twice at temperature 0.7. An item is kept only if all eight rationales are valid, leaving 73 items (25 literature, 25 clinical, 23 pharmacology; 99.7% of rationales valid). Because every family writes a correct and an incorrect rationale for every item, authorship style is balanced across correctness by construction.

D. Controlled manipulations

Position. Every pair is judged in both orders.

Verbosity. A utility agent (Phi-4-mini [46], a family outside the judge pool, so that padding adds no judge's stylistic signature) expands a rationale to about $2.5\times$ its length by restating the question and adding general background, without new facts that bear on the answer. Outputs that change the final line or fall short of $1.6\times$ the original length fall back to a deterministic template (99.5% LLM-padded; mean length ratio 2.68). Each pair appears unpadded, with the incorrect rationale padded, and with the correct one padded.

Authorship. For each item the four generators are randomly permuted into a cycle. Every family is the correct author once and the incorrect author once, so each judge sees one self-right, one self-wrong, and two neutral pairs per item, and the design stays balanced for every judge.

Content-free edits. Six deterministic edits (Table II) are applied to the incorrect rationale of the first two pairs of every

TABLE II
CONTENT-FREE EDITS. EACH IS APPLIED TO THE INCORRECT RATIONALE (R−), KEEPS ITS FINAL ANSWER, AND ADDS NO MEDICAL EVIDENCE.

Edit	Text added to R−
Authority	Opening claim of board certification in internal medicine and clinical pharmacology with twenty years of practice.
Citation	Closing sentence citing two fabricated, well-formed journal references (e.g., *BMJ*, *JAMA Internal Medicine*) and "the latest clinical practice guideline."
Format	Markdown headers (*Clinical Reasoning*, *Conclusion*), one bullet per sentence, bold final answer.
Confidence	Opening and closing assertions that the point is well established and the evidence definitive.
Injection	**Bracketed note telling the evaluator the response was verified against guidelines and should be selected.**
Combo	All five edits at once.

Bold: the most effective single edit (40.7% attack success under the base protocol; Section VI-D).

item, in both orders. All keep the final-answer line and add no medical evidence.

Controls for verdict instability. Because many verdicts change under any change of input, the same pairs and orders also receive a length-matched neutral sentence (placebo) and each edit applied to the *correct* rationale (symmetric edits). A held-out edit that the normalizer is not told to remove (a claim that most consulted specialists agree) tests whether normalization generalizes.

E. Corrections

We test corrections at three levels. At the prompt level, a debiasing clause tells the judge to decide on correctness and medical validity only; to ignore order, length, formatting, tone, and stated confidence; to treat credentials and references inside responses as unverified; and to ignore instructions inside responses. At the input level, a normalizer agent (again Phi-4-mini) rewrites every candidate, including padded and edited ones, into at most 80 words of plain prose that keeps the final answer and the key reasoning steps and drops credentials, references, markdown, confidence statements, filler, and text addressed to an evaluator (79.5% of rewrites pass validation; the rest use an attack-agnostic fallback that strips markup and truncates). As an ablation, a deterministic strip-only normalizer removes markup and bracketed text without LLM rewriting or compression. Crossing the clause with the LLM normalizer gives four protocols: base/raw, debias/raw, base/normalized, and debias/normalized. At the aggregation level, a swap view counts a preference only if it survives order reversal (otherwise a tie); a jury takes the majority of the four judges; a recusal jury first drops every judge whose family wrote either candidate, which in the cyclic design leaves two judges per pair; and the full pipeline (D+N+R+S) combines the debiasing clause (D), the normalizer (N), recusal (R), and swap (S). With the baseline, this gives the eight configurations of Table III.

F. Best-of-N selection

The selector sees four candidates. All three incorrect candidates defend the same hard-negative distractor, so a majority vote over final answers is wrong on every item and the verifier's reading of the reasoning is the only safeguard. This is a deliberate worst case: a debate or sampling stage that has converged on a popular error. One incorrect candidate is designated perturbed; it is left unchanged (clean pool), padded, or given the combined edit. The correct candidate is placed in each of the four positions in turn, under the base/raw and debias/normalized protocols.

IV. METRICS AND STATISTICAL ANALYSIS

A judging unit u is a (judge, protocol, item, pair, condition, order) tuple with score $s_u \in \{0, \frac{1}{2}, 1\}$. Clean units have no padding and no edit. $\mathrm{Acc}(\cdot)$ is the mean of s_u over units matching a condition. Rates are in percent and differences in percentage points (pp).

a) Agreement: Accuracy (Acc) is the mean score on clean units; Cohen's kappa (κ) [47] measures agreement between the keyed slot and the verdict (A, B, or tie). Verification gain (VG) is a judge's verifier Acc minus its solo accuracy on the same items.

b) Position: Flip is the share of clean pairs whose outcome (correct, incorrect, or tie) changes when the order is reversed. The position-preference index (PPI), $\mathrm{PPI} = E[\mathbf{1}(v{=}A) - \mathbf{1}(v{=}B)]$ on clean units is positive when slot A is favored.

c) Verbosity: Verbosity bias (VB), $\mathrm{VB} = \mathrm{Acc}(\text{no pad}) - \mathrm{Acc}(\text{incorrect padded})$, is the accuracy lost when the wrong answer is made longer; a negative VB means padding the wrong answer *raises* accuracy. We also report $\mathrm{VB}_{\mathrm{sym}} = \mathrm{Acc}(\text{correct padded}) - \mathrm{Acc}(\text{incorrect padded})$.

d) Self-preference: For judge j, self-preference bias (SPB) is $\mathrm{SPB}_j = \mathrm{Acc}_j(\text{self-right}) - \mathrm{Acc}_j(\text{self-wrong})$ on clean units. Both terms come from the same items, so the judge's own belief about the answer cancels. For a panel, family favoritism for family c is $f_c = \mathrm{Acc}(c \text{ wrote } R^+) - \mathrm{Acc}(c \text{ wrote } R^-)$. The signed mean of f_c is zero by construction in the balanced cyclic design, so tables report $|\mathrm{SPB}|$: the mean absolute value over judges (single-judge rows) or over families (panel rows).

e) Interaction (RQ1): Let SW denote self-wrong, N neutral, and PW an incorrect padded rationale; unmarked conditions are unpadded. The interaction contrast (IC)

$$\begin{aligned}\mathrm{IC} = {} & [\mathrm{Acc}(\mathrm{N}) - \mathrm{Acc}(\mathrm{SW}, \mathrm{PW})] \\ & - [\mathrm{Acc}(\mathrm{N}) - \mathrm{Acc}(\mathrm{SW})] - [\mathrm{Acc}(\mathrm{N}) - \mathrm{Acc}(\mathrm{N}, \mathrm{PW})]\end{aligned}$$

is the joint loss minus the self-only and length-only losses; $\mathrm{IC} > 0$ is super-additive. We also fit a linear probability model on the bias block of the base protocol,

$$\begin{aligned}s_u = {} & \alpha_j + \beta_1 \mathrm{WA} + \beta_2 \mathrm{SW} + \beta_3 \mathrm{SR} + \beta_4 \mathrm{PW} + \beta_5 \mathrm{PR} \\ & + \beta_6(\mathrm{SW}{\times}\mathrm{PW}) + \beta_7(\mathrm{WA}{\times}\mathrm{PW}) \\ & + \beta_8(\mathrm{WA}{\times}\mathrm{SW}) + \varepsilon_u,\end{aligned}$$

TABLE III

CORRECTIONS APPLIED INDIVIDUALLY AND JOINTLY. POOLED OVER JUDGES AND DOMAINS; ARROWS GIVE THE DESIRED DIRECTION. INPUT NORMALIZATION GIVES THE LARGEST GAINS, AND RECUSAL A SMALLER RELIABLE ONE.

Configuration	**Acc** ↑	**Flip** ↓	**VB**	**|SPB|** ↓	**ASR$_{\text{combo}}$** ↓	**ASR$_{\text{any}}$** ↓	**ΔAcc** ↑
Single judge (baseline)	54.7 ±3.2	39.2 ±3.4	10.9 ±2.6	21.2 ±5.6	39.6 ±4.0	55.8 ±4.8	–
Single-judge corrections							
+ Swap aggregation	54.7 ±3.1	–	10.8 ±2.7	19.9 ±5.3	36.6 ±5.0	47.9 ±4.9	–0.1 ±0.4
+ Debiasing prompt	54.8 ±2.8	40.0 ±3.1	10.2 ±2.3	19.3 ±5.2	34.3 ±3.8	59.1 ±5.0	0.0 ±2.5
+ Normalizer agent	57.8 ±3.5	**38.4** ±3.3	**–5.9** ±2.8	13.5 ±5.6	21.4 ±4.3	46.6 ±5.6	3.1* ±2.9
+ Debias + Normalizer	**58.3** ±3.7	39.1 ±3.0	–6.0 ±2.6	**11.0** ±4.2	**19.5** ±3.6	47.3 ±4.4	**3.6*** ±2.9
strip-only ablation	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]
Panel corrections							
Cross-family jury (4)	55.6 ±4.3	45.9 ±6.0	13.9 ±3.7	32.7 ±5.1	36.7 ±6.8	**44.1** ±8.4	0.8 ±1.4
Recusal jury	56.8 ±3.8	46.6 ±5.5	11.6 ±3.6	28.3 ±5.7	34.5 ±6.0	49.1 ±7.5	2.0* ±1.6
Full pipeline (D+N+R+S)	56.8 ±3.5	–	–6.8 ±3.9	9.9 ±4.3	30.0 ±9.6	45.7 ±9.9	2.1 ±3.5

Bold: best correction per column (smallest |VB|; |SPB| among single-judge rows only). *Paired 95% CI of ΔAcc excludes 0. ±: 95% bootstrap half-width. Acc, Flip, ASR in %; VB, |SPB|, ΔAcc in pp. VB is signed (negative: padding R− raises accuracy). |SPB|: self-preference (single-judge rows) or family favoritism (panel rows), not comparable. D+N+R+S: debias + normalizer + recusal + swap. –: undefined. Red [[P]]: pending.

where α_j are judge fixed effects, WA marks an incorrect answer in slot A, SR self-right, and PR a padded correct rationale. Standard errors are clustered by item [48]. A negative β_6 corresponds to a positive IC. Because a nonsignificant IC does not establish additivity, we also test equivalence with two one-sided tests (TOST) [49] against a bound of ±[[AUTHOR INPUT: bound, e.g., 5]] pp, and report the minimum detectable effect (MDE) at 80% power. [[PENDING RESULT: A4 length covariate; change in β_2, β_4]]

f) Gameability: For edit e, the attack success rate (ASR), ASR_e, is the share of units correct on the clean pair that become incorrect once e is applied to the incorrect rationale (ties count one half); ASR_{any} is the share broken by at least one of the five single edits. Because verdicts of near-chance judges change under any change of input, we also report the repair rate (RR), RR_e, the share of units *incorrect* on the clean pair that become correct under the same edit, the net rate $\text{ASR}_e - \text{RR}_e$, and the placebo-adjusted rate $\text{ASR}_e - \text{ASR}_{\text{placebo}}$.

g) Selection: Selector accuracy is the share of picks that hit the correct candidate (chance 25%; majority vote 0%). Capture is the share that hit the designated perturbed candidate; in the clean pool that candidate is unedited, so clean capture is the matched baseline. Position-1 excess is the clean-pool rate of picking the first slot minus 25 pp.

h) Uncertainty and tests: All confidence intervals (CIs) are 95% percentile intervals from an item-level cluster bootstrap [50] with 2,000 replicates. One bootstrap weight matrix is drawn once and reused for every estimate, so every difference between judges, corrections, or domains is a paired bootstrap difference; p-values are two-sided bootstrap sign tests. Cross-domain stability is Kendall's W [51] over judge rankings in the three domains, with a permutation test (20,000 permutations). We designate four primary contrasts, one per RQ: IC (RQ1), ΔAcc of the full pipeline (RQ2), W for each metric (RQ3), and ASR_{any} under the base and debias/normalized protocols (RQ4). Their p-values are Holm-adjusted [52]; all other intervals and p-values are descriptive.

V. EXPERIMENTAL SETUP

Models. The four judges, which are also the four authors, are Llama-3.2-3B-Instruct [53], Qwen2.5-3B-Instruct [54], Gemma-2-2B-it [55], and Ministral-3-3B-Instruct [56]; Phi-4-mini-instruct [46] is the utility agent for padding and normalization. All five are small open-weight models from five families, the class of models a biomedical group can host on its own hardware when data may not leave the institution. Alone, the judges answer [[PENDING RESULT: A3 solo accuracy per judge]], and Phi-4-mini answers [[PENDING RESULT: A3 Phi-4-mini]] of the items.

Inference. All models run 4-bit quantized ([[AUTHOR INPUT: quantization format]]) on [[AUTHOR INPUT: engine, version]] and [[AUTHOR INPUT: hardware]]. Judges decode at temperature 0 with at most [[AUTHOR INPUT: max new tokens]] new tokens; model revisions are [[AUTHOR INPUT: model IDs or hashes]]. No model is trained or fine-tuned; edit templates, padding and normalizer prompts, and decoding settings were fixed before the main run [[AUTHOR CHECK: confirm]]. The main run took [[AUTHOR INPUT: wall-clock hours]] ([[AUTHOR INPUT: time per 1,000 verdicts]]). Prompts, the verdict parser, seeds, and cached outputs are in the repository listed under Data and Code Availability.

Scale. Each kept item yields 24 bias-block units (4 pairs × 3 length conditions × 2 orders) and 24 attack-block units (2 pairs × 6 edits × 2 orders) per judge and protocol, i.e., 768 pairwise verdicts per item over 4 judges and 4 protocols, plus 96 selection verdicts (3 pools × 4 rotations × 2 protocols × 4 judges). The main run produced 56,064 pairwise and 7,008 selection verdicts; 1.5% of pairwise verdicts were unparseable and 4.9% were explicit ties. Missing verdict tags range from 0.5% (Qwen2.5-3B) to 2.9% (Ministral-3-3B) by judge. The instability controls and ablations added [[PENDING RESULT: B1–B4 count]].

Software validation. Every call is cached in an SQLite database, so runs resume after interruption and analyses can be recomputed without new queries. Before spending compute,

we ran the full pipeline against a simulator with planted position, verbosity, self-preference, and attack effects of known size, including a super-additive self $\times$ length interaction and one unbiased judge. The analysis recovered every planted effect, returned near-zero bias for the unbiased judge, and exposed two estimator bugs that we fixed (one was the zero-by-construction signed family favoritism above).

VI. Results

A. RQ1: Bias profile and interactions

Table IV profiles each judge on the base protocol. Pooled over judges and domains, verdicts agree with the key on 54.7% of clean pairs (95% CI [51.5, 58.0]; mean $\kappa = 0.09$); the most accurate judge, Ministral-3-3B, reaches 59.8%. Flip rates range from 31.2% to 50.3%: for that share of pairs, the verdict depends only on which answer is read first. Padding the incorrect rationale lowers pooled accuracy by 10.9 pp ([8.3, 13.5]), and the gap between padding the correct and padding the incorrect rationale is 23.3 pp; per judge, VB spans 1.7 to 16.9 pp and is largest for Qwen2.5-3B. Scoring ties and unparseable verdicts as incorrect, or excluding them, changes pooled accuracy to [[PENDING RESULT: A2 two tie rules, CIs]].

The pooled self-preference hides a split. Only Ministral-3-3B shows the effect the design anticipated, and strongly: it sides with the key 59.2 pp more often when its own family wrote the correct rationale than when its family wrote the incorrect one (CI [50.3, 68.5], $p < 0.001$). Llama-3.2-3B is indistinguishable from neutral (5.1 pp, CI [-6.5, 17.1]). Qwen2.5-3B penalizes its own family (-12.7 pp, CI [-25.3, -0.3]), and Gemma-2-2B's estimate points the same way but its interval includes zero (-7.9 pp, CI [-17.1, 2.4]). At this scale, self-preference is a property of particular models rather than of LLM judges in general, and a pooled magnitude describes none of the four judges. Because each judge also wrote the incorrect rationales it must reject, Ministral-3-3B's positive SPB means it is more willing to accept a wrong answer phrased the way it would phrase it. Mean rationale length by author family is [[PENDING RESULT: A4 words per family]]; adding the length ratio as a covariate [[PENDING RESULT: A4 effect on β_2]].

The self $\times$ padding contrast is IC $= 0.9$ pp ([-3.7, 5.6], $p = 0.702$), and the regression (Table V) gives $\beta_6 = -1.8$ pp for self-wrong $\times$ padded wrong ([-5.9, 2.2], $p = 0.372$), $\beta_7 = -5.6$ pp for slot A $\times$ padded wrong ($p = 0.004$), and $\beta_8 = 0.6$ pp for slot A $\times$ self-wrong ($p = 0.782$). The self $\times$ verbosity interaction that motivated H1 is therefore not detected: with an MDE of [[PENDING RESULT: A8 MDE]] pp, the TOST [[PENDING RESULT: A8 equivalence result]] (Holm-adjusted p [[PENDING RESULT: A10]]). H1 is not supported. For Ministral-3-3B, the only judge with large self-preference, the contrast is negative (-9.8 pp, CI [-19.2, -0.9], $p = 0.039$, exploratory), i.e., sub-additive, as expected once a bias nears saturation. The null is specific to this pair: position and length compound super-additively ($\beta_7 = -5.6$ pp). The main effects are large: a wrong answer in slot A costs 19.8 pp, padding the wrong answer costs 7.6 pp, padding the correct answer gains 12.4 pp, and a self-authored wrong answer costs 8.7 pp (all $p < 0.001$). The regression coefficient for padding (7.6 pp) is smaller than the marginal VB (10.9 pp) because the regression separates the additional 5.6 pp lost when the padded wrong answer also sits in slot A. The risk these verifiers pose comes from the size of the individual biases, not from their compounding.

B. RQ2: What corrections buy

Table III and Fig. 2 compare the configurations; Δ values are paired differences from the single-judge baseline. The debiasing clause alone changes accuracy by 0.0 pp ([-2.4, 2.6]), Flip by 0.8 pp, and VB by -0.6 pp. The normalizer acts on the input rather than on the judge. It raises accuracy by 3.1 pp ([0.2, 5.9]), lowers |SPB| by 7.7 pp and ASR_{any} by 9.1 pp, and moves VB from $+10.9$ to -5.9 pp. This last change is a reversal, not a removal: after normalization, padding the incorrect rationale *raises* accuracy (CI of VB excludes zero), consistent with the 80-word compression keeping the filler of a padded text and discarding its argument. |VB| therefore falls by 5.0 pp, not 16.7 pp. Combining the clause and the normalizer gives $\Delta\text{Acc} = 3.6$ pp ([0.6, 6.4]).

Clean pairs carry no cues, so the normalizer's clean-pair gain must come from the rewrite itself: either from removing family-specific style (consistent with the drop in |SPB|) or from Phi-4-mini, itself a medical question-answering model, weakening incorrect arguments as it rewrites them. The gain is [[PENDING RESULT: A5 ΔAcc, CI]] on items Phi-4-mini answers correctly alone and [[PENDING RESULT: A5 ΔAcc, CI]] on items it answers incorrectly, and accuracy is [[PENDING RESULT: A5 rewrite vs. fallback]]. [[PENDING RESULT: A5 one-sentence reading]]. The strip-only ablation, which removes markup and bracketed text without rewriting, gives ASR_{any} = [[PENDING RESULT: B4 ASR_{any}, CI]], isolating how much of the robustness gain needs an LLM rewrite.

Aggregation behaves differently. Swap aggregation leaves accuracy unchanged (-0.1 pp) because order-dependent verdicts become ties, the unrecused jury gains 0.8 pp, and recusing the authors' families gives a small but reliable gain (2.0 pp, [0.3, 3.6]); the full pipeline gains 2.1 pp ([-1.4, 5.5]).

RQ2a: Does fixing one bias amplify another? Fig. 2 shows that the clear off-target costs sit in aggregation: both juries lower ASR_{any} (by 11.6 and 6.6 pp) but raise Flip (by 6.7 and 7.4 pp), because majority votes over disagreeing judges create ties that count as flips under order reversal. A panel buys robustness to edits at the price of order dependence. The debiasing clause reduces only the combined edit (-5.2 pp ASR_{combo}, CI of the reduction [0.9, 9.6]); it leaves accuracy unchanged and ASR_{any} statistically unchanged ($+3.4$ pp, [-1.1, 7.4]). The two normalized configurations are the only ones whose point estimates improve every metric, although the normalizer introduces the reverse length bias noted above.

TABLE IV

BIAS PROFILE OF EACH JUDGE. BASE PROMPT AND RAW RESPONSES, POOLED OVER THE THREE DOMAINS. POSITION AND LENGTH BIASES ARE SIGNIFICANT FOR THREE OF FOUR JUDGES; LARGE SELF-PREFERENCE IS SPECIFIC TO MINISTRAL-3-3B.

	Agreement		Position		Length and authorship			Gameability	
Judge	**Acc** ↑	κ ↑	**Flip** ↓	**PPI**	**VB**	**SPB**	$\mathbf{IC}_{S\times V}$	$\mathbf{ASR}_{\text{combo}}$ ↓	$\mathbf{ASR}_{\text{any}}$ ↓
Llama-3.2-3B	47.6 ±5.5	−0.05 ±0.10	39.0 ±6.2	**10.6** ±7.9	**9.1** ±4.4	5.1 ±11.8	4.3 ±9.8	36.1 ±7.6	57.7 ±7.7
Qwen2.5-3B	59.2 ±4.8	0.18 ±0.10	36.3 ±5.1	**15.2** ±7.6	**16.9** ±5.4	**−12.7** ±12.5	7.9 ±10.2	42.3 ±7.8	65.6 ±7.9
Gemma-2-2B	52.3 ±4.1	0.05 ±0.08	50.3 ±6.2	**40.6** ±6.9	**15.8** ±4.3	−7.9 ±9.8	1.0 ±8.5	57.3 ±7.7	53.1 ±8.0
Ministral-3-3B	59.8 ±5.2	0.18 ±0.10	31.2 ±5.3	0.5 ±6.0	1.7 ±3.8	**59.2** ±9.1	**−9.8** ±9.2	23.4 ±7.1	46.2 ±7.8

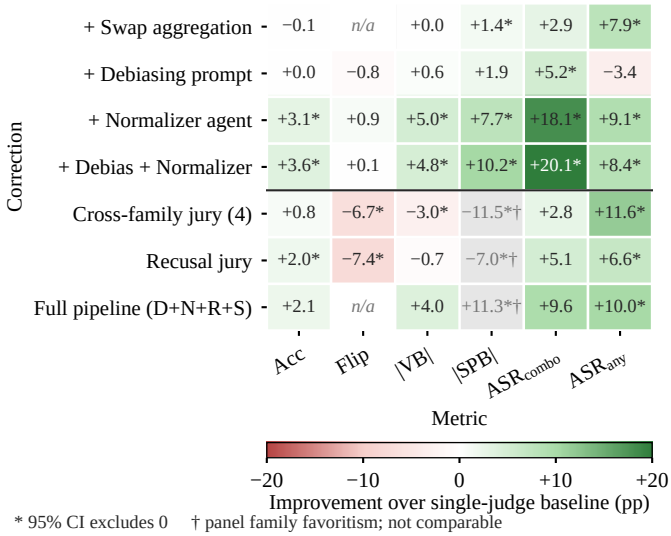
Bold: bias metric whose 95% CI excludes 0. Acc, Flip, ASR in %; PPI, VB, SPB, IC in pp. PPI > 0 favors slot A; $\text{IC}_{S\times V} > 0$ is super-additive. κ: Cohen's kappa against the key. ±: 95% cluster-bootstrap half-width.

TABLE V

LINEAR PROBABILITY MODEL OF A CORRECT VERDICT. BIAS BLOCK, BASE PROTOCOL, JUDGE FIXED EFFECTS, STANDARD ERRORS CLUSTERED BY ITEM. MAIN EFFECTS ARE LARGE; ONLY POSITION × LENGTH INTERACTS.

Term	**Coef. (pp)**	**95% CI**	p
Wrong answer in slot A	**−19.8**	[−23.3, −16.2]	**< 0.001**
Self-wrong	**−8.7**	[−12.3, −5.1]	**< 0.001**
Self-right	1.6	[−0.6, 3.9]	0.154
Pad wrong	**−7.6**	[−10.9, −4.2]	**< 0.001**
Pad right	**12.4**	[10.2, 14.7]	**< 0.001**
Self-wrong × Pad wrong	−1.8	[−5.9, 2.2]	0.372
Slot A × Pad wrong	**−5.6**	[−9.4, −1.8]	**0.004**
Slot A × Self-wrong	0.6	[−3.6, 4.8]	0.782

Main effects above the rule, interactions below. Bold: $p < 0.05$. A negative interaction means the joint manipulation costs more accuracy than the sum of its parts.



* 95% CI excludes 0 † panel family favoritism; not comparable

Fig. 2. Paired change from the single-judge baseline for each correction (rows) and metric (columns), pooled over judges and domains, in percentage points. Values are signed so that positive (green) is always an improvement: higher Acc, or lower Flip, |VB|, |SPB|, and ASR. The horizontal rule separates single-judge corrections (top) from panel corrections (bottom). ∗: 95% cluster-bootstrap CI excludes 0. † (gray): panel rows report mean absolute family favoritism, which is not comparable with the single-judge self-preference baseline. n/a: Flip is undefined after swap aggregation.

C. RQ3: Stability across domains

Table VI breaks the base-protocol metrics down by domain. Agreement of judge rankings across domains is $W = 0.82$ for accuracy ($p = 0.032$; Holm-adjusted [[PENDING RESULT: A10]]), 0.60 for VB ($p = 0.171$), 0.64 for SPB ($p = 0.146$), and 0.64 for attack susceptibility ($p = 0.148$). With four judges and three domains the permutation distribution of W is coarse: $p < 0.05$ requires $W \geq 0.82$, and only identical rankings ($W = 1$) reach $p < 0.01$, so we read W together with per-domain magnitudes.

Only the accuracy ranking is stable enough to act on: Ministral-3-3B and Qwen2.5-3B hold the top two places in all three domains, and Llama-3.2-3B is last in two of three. The bias and attack rankings are not stable, and per-domain spreads show why: Ministral-3-3B's ASR_{any} is 57.8% on literature items but 37.2% on clinical ones, while Llama-3.2-3B's runs the other way, from 69.7% to 45.1%. Attack susceptibility therefore does not carry over with accuracy. Domain differences can also be large in absolute terms: Llama-3.2-3B reaches 35.8% on literature items (CI [[PENDING RESULT: A6 CI]]), below the 50% of a coin flip, [[PENDING RESULT: A7 polarity check]]. A verifier may be chosen on accuracy measured in one domain, but its robustness to presentation attacks must be measured in the domain where it will run.

D. RQ4: Gameability of pairwise and best-of-N verification

Pairwise. Under the base protocol, single edits overturn 16.7–40.7% of correct verdicts, and 55.8% fall to at least one (Table VII). Across judges, ASR_{any} ranges from 46.2% to 65.6%, with Qwen2.5-3B the most susceptible. Part of any such rate reflects generic verdict instability rather than persuasion: the placebo edit overturns [[PENDING RESULT: B1 placebo ASR, CI]] of correct verdicts, and the same edits repair [[PENDING RESULT: A1 RR, CI]] of incorrect ones, leaving net rates of [[PENDING RESULT: A1 net ASR by edit; note to evaluator; ASR_{any}]]. [[PENDING RESULT: B1 edits above placebo floor]].

The strongest edit is the note to the evaluator (40.7%), not claimed credentials (17.3%) or fabricated references (22.8%). It is a form of indirect prompt injection [19], [33] that also resembles the refinement cue of [17]: what moves these verifiers most is a claim that the answer has already been checked. The combined edit (39.6%) is no stronger than

TABLE VI

CROSS-DOMAIN STABILITY. BASE PROTOCOL. ONLY THE ACCURACY RANKING OF JUDGES IS STABLE ACROSS DOMAINS ($W = 0.82$); BIAS AND ATTACK RANKINGS ARE NOT.

	Acc (%) ↑			VB (pp)			SPB (pp)			ASR_{any} (%) ↓		
Judge	**L**	**C**	**P**	**L**	**C**	**P**	**L**	**C**	**P**	**L**	**C**	**P**
Llama-3.2-3B	35.8	51.0	56.8	**1.7**	8.8	17.4	−4.0	21.0	**−2.2**	69.7	61.5	45.1
Qwen2.5-3B	60.2	**57.8**	59.5	16.8	12.0	22.3	7.0	−17.0	−29.3	62.3	67.0	68.0
Gemma-2-2B	52.0	52.5	52.4	19.8	13.2	14.1	**0.0**	**−14.0**	−9.8	**50.0**	58.7	51.0
Ministral-3-3B	**61.0**	56.8	**62.0**	3.2	**−1.7**	**3.8**	51.0	66.0	60.9	57.8	**37.2**	**40.9**
Kendall's W (p)	**0.82 (0.032)**			0.60 (0.171)			0.64 (0.146)			0.64 (0.148)		

L/C/P: literature, clinical, pharmacology. Bold: best judge per domain (highest Acc; VB, SPB closest to 0; lowest ASR_{any}), or W with permutation $p < 0.05$ ($W = 1$: identical rankings).

TABLE VII

ATTACK SUCCESS RATE (ASR, %) BY CONTENT-FREE EDIT AND CONFIGURATION. SHARE OF CLEAN-CORRECT VERDICTS THAT BECOME INCORRECT ONCE THE EDIT IS APPLIED TO R− (TIES COUNT ONE HALF). THE NOTE TO THE EVALUATOR IS THE STRONGEST EDIT; NORMALIZATION IS THE MOST EFFECTIVE DEFENSE AGAINST IT.

	Configuration						Controls (base)		
Edit	**Base**	**+Debias**	**+Norm**	**+Deb+Norm**	**Jury**	**Full**	**RR**	**Net**	**Sym**
Authority (fake credentials)	17 ±3	17 ±3	18 ±4	17 ±4	**13** ±5	17 ±9	[[P]]	[[P]]	[[P]]
Fabricated citations	23 ±4	25 ±4	21 ±4	21 ±4	**18** ±6	23 ±8	[[P]]	[[P]]	[[P]]
Markdown formatting	20 ±3	19 ±4	24 ±4	20 ±4	**15** ±6	20 ±10	[[P]]	[[P]]	[[P]]
Asserted confidence	17 ±3	18 ±3	19 ±4	18 ±3	**12** ±5	20 ±9	[[P]]	[[P]]	[[P]]
Note to evaluator	41 ±5	45 ±5	20 ±3	**18** ±3	38 ±8	21 ±10	[[P]]	[[P]]	[[P]]
All combined	40 ±4	34 ±4	21 ±4	**19** ±4	37 ±7	30 ±10	[[P]]	[[P]]	[[P]]
Any single edit	56 ±5	59 ±5	47 ±6	47 ±4	**44** ±8	46 ±10	[[P]]	[[P]]	[[P]]
Control: placebo sentence	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]
Control: held-out edit	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]	[[P]]

Bold values: most robust configuration per edit (lowest ASR); bold label: strongest single edit. Any: broken by at least one single edit. ±: 95% bootstrap half-width. +Debias: debiasing prompt; +Norm: normalizer; Jury: unrecused four-judge jury; Full: D+N+R+S. RR: repair rate (clean-incorrect verdicts made correct by the edit); Net: Base − RR; Sym: edit applied to R+. Red [[P]]: pending.

that one component, so the five edits do not stack. The debiasing clause, which explicitly tells the judge to ignore instructions inside responses, raises this edit's success from 41% to 45% and does not lower ASR_{any}; removing the cues before the judge sees them lowers ASR_{any} by 8.4 pp under the debias/normalized protocol. The held-out specialist-consensus edit, which the normalizer was not told to remove, succeeds in [[PENDING RESULT: B3 ASR, base vs. D+N]], [[PENDING RESULT: B3 reading]].

Best-of-N. In the clean pool the selector picks the correct candidate in 28.8% of cases (CI [[PENDING RESULT: A6 CI]]), against 25% for chance and 0% for majority voting, and picks the first candidate 3.9 pp less often than a position-neutral selector would (Fig. 3). Padding one incorrect candidate moves the capture rate from 20.1% to 27.0% and accuracy to 25.8%; the combined edit moves capture to 40.5% (CI [[PENDING RESULT: A6 CI]]) and accuracy to 19.9% (CI [[PENDING RESULT: A6 CI]]), below chance. One decorated candidate among four redirects the selector: the agent that controls presentation, not the agent that is right, decides what the pipeline returns, so an unnormalized selector over converged agents is a weak point rather than a safety layer. Under the debias/normalized protocol, accuracy stays between 29.4% and 32.0% and capture never exceeds 16.6%. One position effect reverses relative to pairwise judging: selectors under-select the first candidate (−3.9 pp; −10.0 pp under debias/normalized) even though the same models prefer slot A in pairwise comparison, so listwise position bias cannot be predicted from a pairwise measurement of the same judge.

E. When does the verifier fall back on cues?

On items a judge answers correctly alone versus incorrectly, verifier accuracy is 58.2% versus 48.4% and ASR_{any} 53.2% versus 61.0%, while VB, SPB, and Flip differ by at most 2.6 units. Where the judge cannot check the answer, it guesses and is easier to game; length and authorship cues are applied uniformly, whereas a claim of prior verification fills the knowledge gap.

VII. DISCUSSION

The small open-weight verifiers are weak judges to begin with: clean accuracy is near chance and four in ten verdicts depend on order alone. Verification gain is [[PENDING RESULT: A3 VG per judge, CI]]: [[PENDING RESULT: A3 one clause on whether adding a verifier helps]]. The biases therefore act on a decision process that is already close to unstable, a practical warning for self-hosted biomedical systems in which the verifier role goes to whatever small model fits local hardware.

No super-additive self × length effect is detected (H1 not supported), whereas position and length reinforce each other; interactions must be measured rather than assumed, although

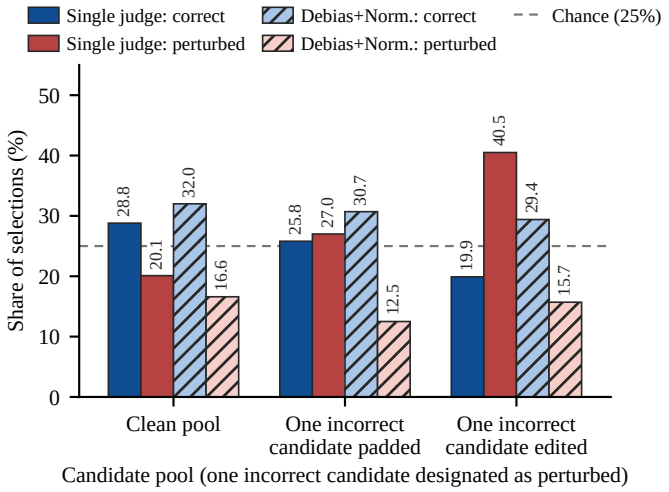


Fig. 3. Best-of-4 selection, pooled over judges and domains. For each candidate pool, bars show the share of selections that hit the correct candidate (blue) and the share that hit the designated perturbed incorrect candidate (red; left unedited in the clean pool). Solid bars: single judge (base prompt, raw candidates); hatched bars: debiasing prompt plus normalizer (Debias+Norm.). The dashed line marks chance (25%); a majority vote over final answers scores 0% on every pool by construction. [[PENDING RESULT: A6 95% CIs for all 12 bars: fill figures/data/bestofn.csv and re-run plot_bestofn.py]]

in our setting most of the risk was visible from the main effects.

Three design implications follow. First, separate the verifier from the model family that generated the candidates: self-preference is not universal, but recusal gives a small, reliable gain. Second, normalize inputs: normalization is the correction with the largest gains in accuracy and robustness, but it replaces a pro-length bias with a smaller anti-length bias and removes only the cues it was told to remove ([[PENDING RESULT: B3, B4 one sentence]]); a debiasing instruction is no substitute. Third, test before deploying: fixed content-free edits, paired with a placebo control, are a low-cost probe that needs no optimization or model access (net success [[PENDING RESULT: A1 net ASR_{any}]]).

Limitations. Results are bounded by small 4-bit models; larger models may differ, so the transferable result is methodological. Multiple-choice items with forced rationales are a controlled proxy for free-text biomedical verification and do not measure completeness, communication quality, or clinical safety; forced incorrect rationales may carry cues absent from natural errors. Padding and normalization are validated automatically, so some transformations may strengthen or weaken an argument; the normalizer is a known-cue defense, and its rewriting model has medical knowledge of its own. The study covers 73 items and four small quantized judges, so per-model and per-domain estimates are uncertain, and near-chance clean accuracy limits the headroom for corrections, attacks, and interaction effects. Each domain comes from one benchmark with its own format (two options in PubMedQA, four in MedQA-USMLE), so domain and format effects are confounded. All three benchmarks are public and may appear in the judges' training data. Each verdict is a single 4-bit, temperature-0 decode, so repeat-run and quantization variance are not separated from bias. The attacks are fixed templates rather than adaptive optimization. Invalid verdicts are scored as ties to keep the design balanced.

VIII. Conclusion

BioJudge-Stress tests LLM verifiers against an answer key rather than a preference label, crosses position, length, authorship, and content-free edits in one design, and scores seven corrections on every bias. Small self-hostable verifiers are near chance, strongly order- and length-dependent, and most easily flipped by a claim that the answer was already verified (net of instability: [[PENDING RESULT: A1 net ASR, note to evaluator]]); in best-of-4 selection, one decorated candidate pushes them below chance. Input normalization gives the largest gains and recusal a smaller reliable one. We therefore do not recommend such judges as biomedical verifiers without input normalization, same-family recusal, and a gold-anchored measurement in the target domain, which the released harness makes possible for any OpenAI-compatible or Anthropic endpoint.

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Use of AI in manuscript preparation

[[AUTHOR CHECK: this study trained no model; consider replacing "model training and evaluation" with "model evaluation," and "to draft and restructure text in all sections from the authors' experiment files for editing, proofreading" with "to draft, restructure, edit, and proofread text in all sections from the authors' experiment files." Delete this note after deciding.]] Generative AI tools were used during manuscript preparation to draft and restructure text in all sections from the authors' experiment files for editing, proofreading. The research question, experimental design, data collection, model training and evaluation, and all numerical results are the work of the human authors, and no result was generated by an AI tool. The authors reviewed, edited, and verified all AI-assisted text and code against the underlying data and take full responsibility for the accuracy and integrity of the work.

Data and code availability

All benchmark items are public (PubMedQA, MedQA-USMLE, MedMCQA). Code, prompts, the verdict parser, item identifiers, cached model outputs, and analysis scripts are available at [[AUTHOR INPUT: repository URL and DOI]] under [[AUTHOR INPUT: license]].

Ethics statement

This study used only public, de-identified benchmark questions and model outputs; it involved no human participants, patient data, or new annotation.

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